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PROTEIN ENGINEERING

Structure and evolution-guided design of minimal RNA-guided nucleases

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The design of RNA-guided nucleases with properties not limited by evolution can expand programmable genome-editing capabilities. However, generating diverse multidomain proteins with robust enzymatic properties remains challenging. Here, we use a protein design strategy that couples a structure-guided inverse-folding model with evolution-informed residue constraints to generate active, divergent variants of TnpB, a minimal CRISPR-Cas12–like nuclease, termed SynTnpBs. High-throughput screening of artificial intelligence–generated variants yielded editors that retained or exceeded wild-type activity in bacterial, plant, and human cells. Cryo–electron microscopy–based structure determination of the most divergent variant revealed stabilizing contacts in the RNA–DNA interfaces across conformations, demonstrating the design potential of this approach. Together, these results establish a strategy for creating non-natural RNA-guided nucleases and conformationally active nucleic acid binders, enlarging the designable protein space.

CRISPR-Cas systems have revolutionized genome editing by enabling programmable, sequence-specific DNA and RNA targeting, forming a foundation for precise genetic and epigenetic perturbations (*1*–*7*). Protein design has the potential to extend these capabilities by creating RNA-guided nucleases with functions and properties, including sequence and structures, that are not observed in nature.

However, design of this type is particularly challenging because these nucleases are multidomain nucleic acid binders whose activity depends on coordinated RNA and DNA recognition, activation, and cleavage by distinct conformational states (*8*–*11*). Sequence-based biological language models (LMs) trained on evolutionary data can generate RNA-guided nucleases by inferring sequence–function relationships, but the active nucleases they produce often remain highly similar to the reference sequences they are trained on, even after extensive post hoc filtering (*12*, *13*). Structure-guided rational design approaches offer a robust strategy to sample highly divergent protein sequences, and also structures not found in nature, extending into de novo design (*14*–*19*). Although this approach has been successful for the generation of dynamic switches and DNA binders (*20*–*22*), the design of complex enzymes, such as RNA-guided nucleases bearing multiple functional domains and conformational states, remains an attractive but unresolved challenge.

We reasoned that inverse protein-folding models (*17*–*19*), a structure-conditioned approach, coupled with evolutionary information could generate highly diverse proteins while retaining function. To test the potential of such a design strategy, we selected TnpB, a diverse family of small transposon-encoded ancestors of CRISPR-Cas12 nucleases that mediate RNA-guided DNA cleavage, transcriptional regulation, and genome targeting across all kingdoms of life (*23*–*29*). These features render TnpBs as a promising scaffold for exploring RNA-guided nucleases not produced by natural evolution.

Evolution- and structure-conditioned inverse folding of TnpB using ESM-IF1

We began our study by evaluating in silico whether the ESM Inverse Folding (ESM-IF1) model (*18*) could redesign the minimal RNA-guided nuclease ISDra2 TnpB to create functional enzymes with non-natural sequences. ESM-IF1–generated sequences accurately recapitulated the input fold predicted by AlphaFold2 (*30*) (Fig. 1A and fig. S1A), preserved the DED catalytic triad in the RuvC domain (Fig. 1B and fig. S1B), and captured amino acid variation found among natural homologs (fig. S1, C and D) (*18*). By contrast, LigandMPNN (*19*, *31*) reproduced the TnpB fold but failed to maintain the catalytic triad across all model generation temperatures.

Phylogenetic analysis placed ESM-IF1 sequences in a distinct clade from that of natural TnpBs (Fig. 1C), sharing only 50 to 60% identity with ISDra2 TnpB, basal to this clade (fig. S1, E and F). The model produced chemically conservative substitutions across the DNA–RNA interface (Fig. 1B). However, it also introduced nonsynonymous mutations at residues K76, F77, and T123, known to mediate nucleobase-specific recognition of the transposon-associated motif (TAM) sequence (*32*), a short DNA motif priming TnpB activation similar to the protospacer adjacent motif (PAM) of CRISPR-Cas9 (*1*). Similarly, residue substitutions diverging from the wild-type (WT) ISDra2 TnpB occurred at the interface between the protein and its guide RNA (or reRNA for TnpB). This suggests that ESM-IF1 captures TnpB's fold and catalytic DED motif and partially recovers nucleic acid–binding residues, but misassigns some contact positions in the absence of a cognate RNA–DNA context.

We therefore explored targeted conditioning to enforce functional protein-RNA and protein-DNA contacts. Typically, inverse-folding models fix residues within a defined radius of experimentally resolved bound ligands (*33*). For TnpB, however, the 49-kDa RNA–DNA heteroduplex binds the 46-kDa protein over a large interface, requiring many residues to be fixed and leaving little space to redesign. Furthermore, for dynamic proteins such as RNA-guided nucleases, available cryo–electron microscopy (cryo-EM) structures (*32*, *34*) may miss transient protein-DNA/RNA contacts, which may result in not fixing all residues necessary for function.

To address this issue, we developed a masking strategy for inverse folding that introduced functionally critical residues from evolutionary data. Pairwise evolutionary couplings derived from Potts models in physics have been foundational for inferring residue contacts, guiding protein structure prediction, predicting protein-protein contacts, and estimating mutational effects (*35*–*38*). Previous bioinformatic studies showed that DNA and RNA binding residues can be identified from

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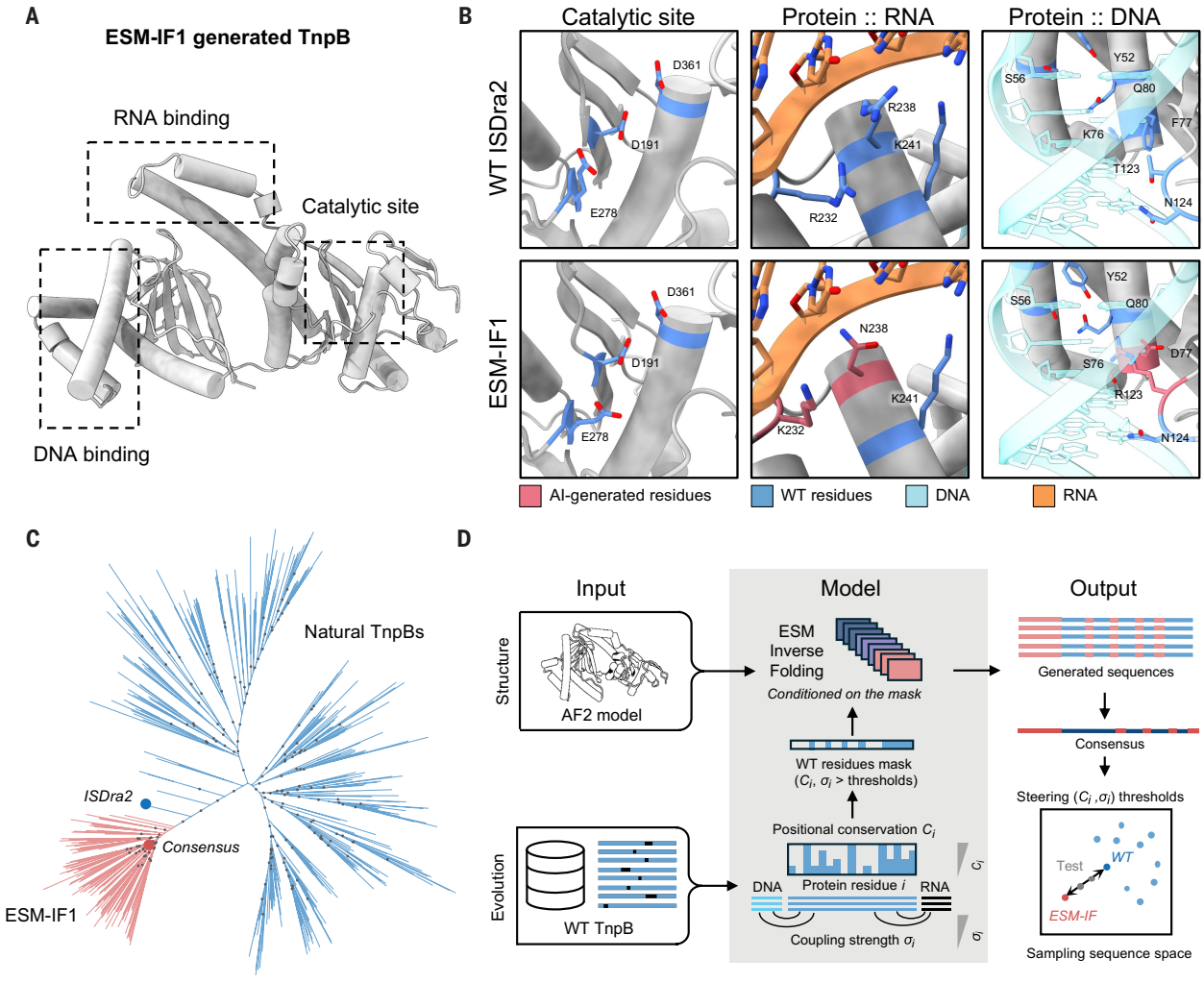


Fig. 1. RNA-guided nuclease TnpB design strategy with an ESM Inverse Folding (ESM-IF1) model. (**A**) ESM-IF1 generates sequences that consistently fold into ISDra2 TnpB structure via AlphaFold2 (*30*) (**B**) RuvC catalytic residues returned by ESM-IF are similar to the WT, whereas those that bind to the nucleic acids vary between WT and ESM-IF sequences. (**C**) Phylogenetic analysis shows that sequences generated by ESM-IF1 cluster into a distinct clade, reflecting substantial divergence from natural TnpBs. Nodes with ≥90% bootstrap support are marked with gray dots. (**D**) To preserve ISDra2 TnpB-specific nucleic acid–binding residues in the generated sequences, the ESM-IF model is conditioned on WT residues conserved above a chosen threshold for positional conservation C_i or coevolutionary coupling strength σ_i to the ligand nucleic acids. By increasing the (C_i, σ_i) thresholds, the divergence from the WT ISDra2 TnpB is gradually increased, ultimately yielding fully unconstrained sequences. Consensus sequences at selected parameters are then calculated and tested for activity.

phylogenetic conservation or evolutionary couplings (*39*, *40*). We thus derived positional conservation C_i values for TnpB residues i from multiple-sequence alignments of natural TnpBs, and coupling signals σ_i from a Potts model (GREMLIN) (*37*) trained on either paired TnpB–RNA or TnpB–DNA sequences mined from genomic databases (materials and methods and fig. S2). Residues exceeding given C_i, σ_i thresholds C_0, σ_0 were fixed to their WT identities in ISDra2 TnpB, producing sequence masks for conditional ESM-IF1 generation. Varying thresholds C_0, σ_0 effectively tuned the balance between structural and evolutionary input to the model (Fig. 1D). We further found that the consensus averaging of generated sequences had lower model perplexity than individual sequences, indicating improved fold-compatibility and possibly higher chances of activity (fig. S3).

High-throughput bacterial screens identify active variants

Our design objective was to experimentally determine which (C_0, σ_0) combinations produced active nucleases while minimizing fixed residues, thereby maximizing the sampled sequence space and exploring synthetic TnpB diversity. To maximize the generation rate of active proteins, we tested consensus sequences derived from 10,000 ESM-IF1 generations per C_0 or dual (C_0, σ_0) conditions. To screen the generated TnpB variants for activity, we used a bacterial assay in which cell recovery is proportional to TnpB-mediated cleavage of a plasmid carrying the *ccdB* toxin gene (*41*) (Fig. 2, A to C).

Using spot plating to test full-length proteins generated for a range of conservation (C_i) thresholds (C_0) and model temperatures (T), we observed that three variants at $C_0 = 0.25$ and 0.35 exhibited activity, with survival levels of 10^3 to 10^4 colony-forming units (CFU)/μg, albeit lower than the 10^6 CFU/μg for the WT ($C_0 = 0.0$; Fig. 2D). We hypothesized that as TnpBs are multidomain proteins, the different lobes may not tolerate comparable mutational depth or achieve similar generation success. Therefore, we implemented a split-domain strategy in which the generated DNA and RNA binding domains, referred to as REC and NUC respectively, were experimentally tested separately.

Inspired by the domain-swapping strategy previously applied to CRISPR-Cas9 engineering (*42*), we fused the artificial intelligence (AI)–generated lobes to their WT counterparts to assess them individually (Fig. 2, E and F). Supporting our hypothesis, the lobes exhibited different tolerances to sequence divergence. Only one AI-generated REC-lobe variant at $C_0 = 0.25$ was functional when fused to the WT

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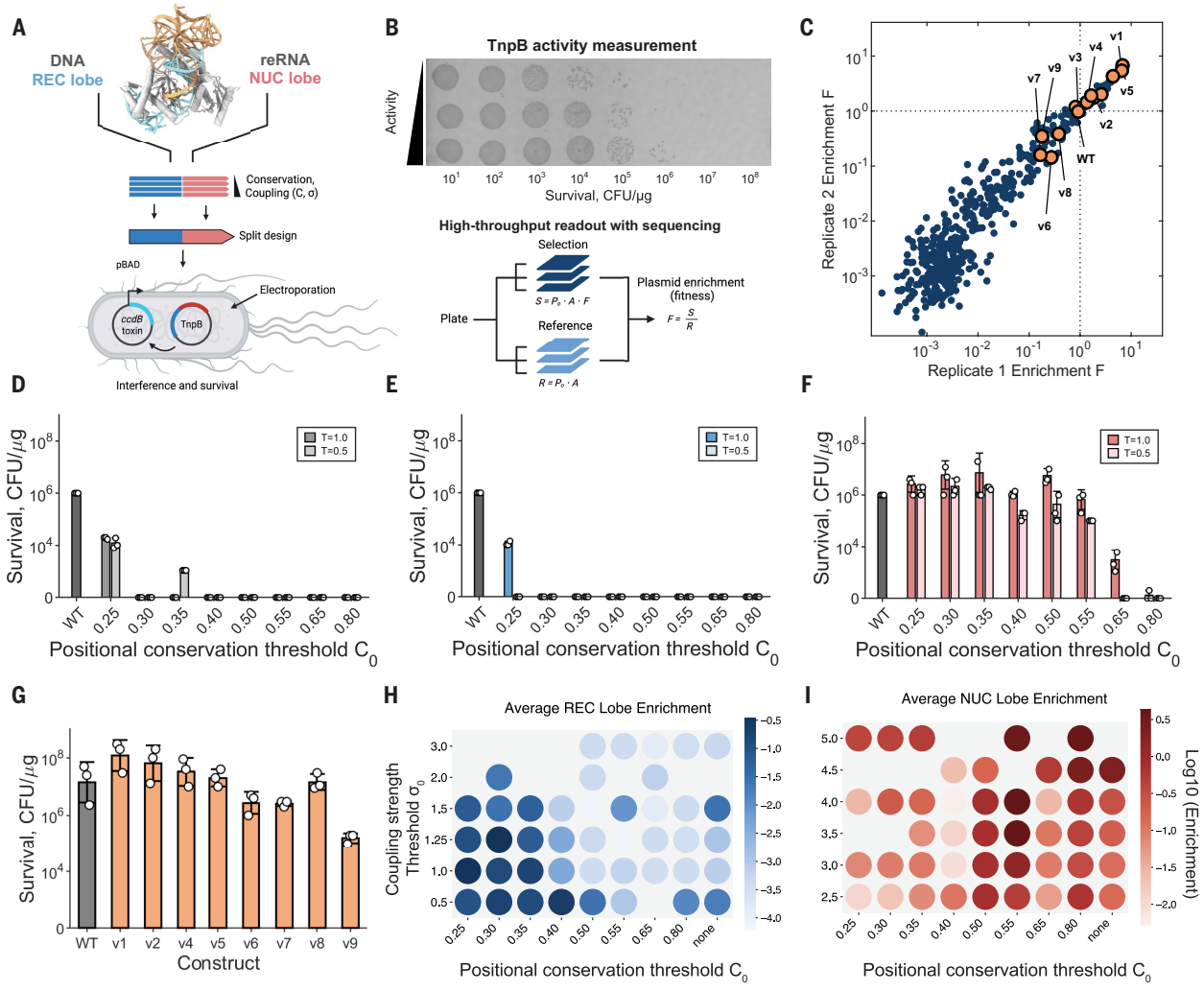


Fig. 2. Screening AI-generated SynTnpB variants conditioned on positional conservation (C_i) and coupling strength (σ_i) thresholds. (**A**) Split-lobe design of TnpB (depicted with 8EXA model) allowed combinatorial testing of REC and NUC lobes. Plasmids expressing TnpB variants were electroporated into *E. coli* cells carrying an arabinose-inducible toxin (pBAD-ccdB). Active variants cleave the toxin plasmid, enabling survival on selective agar. (**B**) Activity was assessed by spot plating (top) or by pooled bacterial selection coupled to high-throughput sequencing (bottom). The equation terms denote the abundance of each generated variant expression plasmid in the initial transformed library (P_0), selection plate (S), and nonselective plate (R). A denotes transformation and growth bias, and F selection fitness associated with toxin-plasmid cleavage. (**C**) Log_{10} enrichment of generated variants across two experimental replicates. Data were normalized such that reference WT ISDra2 TnpB enrichment = 1. Enrichment versus identity to WT is depicted in fig. S4. (**D**) Activities of full-length variants generated across C_i thresholds at two ESM-IF1 sampling temperatures. Bars show mean ± SD from three replicates. (**E** and **F**) Cleavage activities of REC (E) and NUC (F) lobes tested individually with their WT counterparts. (**G**) Recovery of top variants [1980 tested in (C)] assembled from REC and NUC lobes conditioned on both (C_i, σ_i) thresholds. (**H** and **I**) Heat maps showing enrichment of AI-generated REC (H) and NUC (I) lobes over (C_i, σ_i) thresholds, averaged over all fused combinations.

NUC lobe, whereas 13 out of 16 NUC-lobe variants retained activity, often higher than that of the WT, for C_0 thresholds with values as high as 0.65. These results suggest that the NUC lobe accepts higher levels of divergence from the WT, whereas the REC lobe is more sensitive to substitutions under single-parameter (C_i) conditioning.

To sample a broader range of substitutions and further increase the diversity of active variants, we tested a dual-conditioning strategy by fixing residues according to both positional conservation (C_i) and protein–nucleobase coupling strength (σ_i). REC and NUC lobes were independently generated under (C_0, σ_0) thresholds (materials and methods; figs. S2, S4, and S5) and assayed in all pairwise REC–NUC combinations (44 × 45 lobes; 1980 fusions) using a high-throughput pooled library selection assay (Fig. 2B). Activity was quantified as the enrichment of lobe-pair plasmids under selective (+arabinose) versus nonselective (–arabinose) conditions.

In contrast to single-parameter (C_i) conditioning, dual (C_i, σ_i) masking produced a markedly broader distribution of active variants, revealing more diverse and functional REC and NUC lobes (Fig. 2, H and I). Of the 466 lobe combinations that had detectable enrichment in bacterial assays (24% of 1980 designs), ~8% exceeded the activity of the WT (Fig. 2C and fig. S6). The most diverse and active variants identified in the pooled assay (SynTnpBs-v1 to v9) were validated by spot plating (Fig. 2G) and were shortlisted for testing in human and plant genome editing assays.

AI-designed SynTnpB enzymes are active genome editors in human and plant cells

We next evaluated the genome-editing activity of representative SynTnpB variants (v1 to v9), spanning high activity and sequence diversity ranging from 77 to 91%, and WT ISDra2 in an endogenous blue fluorescent protein (*BFP*) gene knockout assay in human embryonic kidney 293T (HEK293T) cells using biological triplicates (Fig. 3A and fig. S7). The WT enzyme exhibited an average of 28% editing. Variants v2 to v4 and v6 to v8 showed values comparable to those of the WT in

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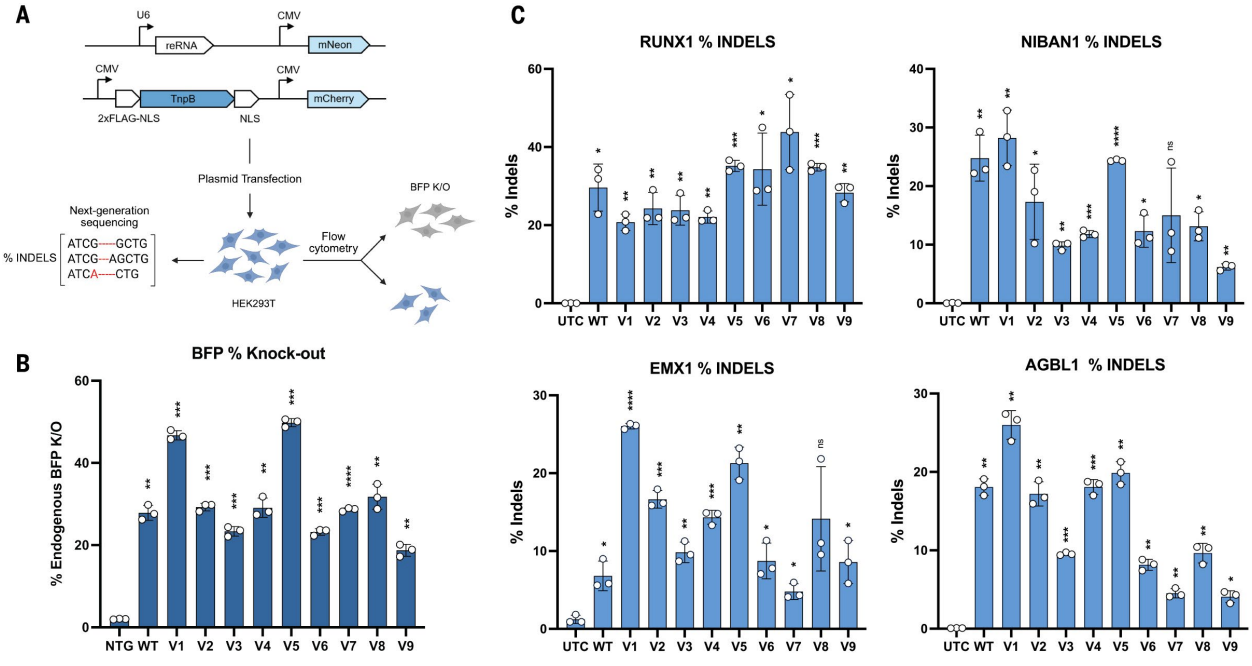


Fig. 3. AI-generated TnpB genome editing in HEK293T cells. (**A**) Plasmids encoding variant proteins and corresponding guide reRNAs were cotransfected into HEK293T cells. Editing efficiency was assessed by either (**B**) *BFP* gene knockout measured by flow cytometry or (**C**) NGS quantification of indels at endogenous loci (*RUNX1*, *NIBAN1*, *EMX1*, *AGBL1*). Statistical significance was determined using a two-tailed Welch's *t* test. Data are presented as mean ± SD ($n = 3$). ns, not significant; *P* value: ns > 0.05, *$P \leq 0.05$, **$P \leq 0.01$, ***$P \leq 0.001$, ****$P \leq 0.0001$. UTC: untransfected control; NTG; nontargeting guide.

the range of 23 to 32%. Notably, v1 and v5 showed the highest increases in editing efficiencies over WT, with values of 46% ($P < 0.001$) and 50% ($P < 0.001$), respectively (Fig. 3B). These results confirmed the successful generation of active and diverse variants.

Next, to assess programmability and editing potential, we targeted endogenous human and plant genomic loci. Four endogenous genes interrogated previously, *RUNX1*, *NIBAN1*, *EMX1*, and *AGBL1*, were targeted by the AI-generated TnpBs in HEK293T cells (*23*, *43*), and the percentage of insertions and deletions (indels) was calculated by next-generation sequencing (NGS). Although most variants exhibited similar activity to that of the WT, v1 and v5 again were the most active variants across the tested loci. Variant v1 showed editing of 26, 26, 28 and 21% for *EMX1*, *AGBL1*, *NIBAN1*, and *RUNX1*, respectively, while v5 showed 21, 20, 24, and 35%, compared to 7, 18, 25 and 30% for the WT (Fig. 3C). Notably, v1 and v5 showed respective 3.8- and 3.1-fold increases over WT at the *EMX1* locus (v1 $P < 0.0001$, v5 $P < 0.01$). No significant differences in editing windows were observed between the AI-designed variants and WT ISDra2 TnpB (fig. S8).

We next assessed the specificity of the AI-designed variants using genome-wide Tn5 tagmentation profiling (materials and methods). Variant v1 showed comparable specificity to that of the WT, whereas v5 and v7 had more detectable off-target sites. All variants retained the canonical TTGAT TAM preference (figs. S9 and S10). Western blot analysis in HEK293T and *Escherichia coli* showed that expression differences across variants do not correlate with editing activity (fig. S11A). Biochemical characterization using in vitro cleavage assays revealed that v7 retains robust target strand cleavage activity, albeit with moderately reduced kinetics relative to WT (fig. S11B). Circular dichroism–based thermal denaturation analysis showed that v7 exhibits comparable thermal stability to that of the WT, with a melting temperature of 51.2°C (fig. S12) (*32*).

Because of their minimal size, TnpBs are well suited for plant genome editing by viral vector delivery (*44*). Thus, we tested the nine selected variants in *Arabidopsis* protoplast cells by targeting the *AtPDS3* gene at four distinct sites. Consistent with the HEK293T results, v1 outperformed ISDra2 at nearly all tested targets (fig. S13 and table S1).

Conformational dynamics provide mechanistic insights into SynTnpB activity

The highly divergent variant v7 showed the highest absolute editing efficiency in our cellular experiments (44% at *RUNX1*; Fig. 3C). This variant shared 77% sequence identity with the WT (excluding the unstructured C terminus dispensable for activity, which was kept fixed; 83% for the REC lobe, 72% for the NUC lobe). In line with our design objective, approximately one in four residues in v7 was generated by the evolution-conditioned ESM-IF1 model, totaling 85 positions. Many of these residues lined the nucleic acid interface alongside masked WT residues and included both conservative and nonconservative substitutions (fig. S14). These results prompted us to examine the structural basis of how AI-introduced substitutions facilitate the SynTnpB cleavage mechanism. To study how these residues facilitate function in the AI-generated protein v7, we determined its ternary complex structure with cryo-EM, capturing TAM-bound and R-loop–formed states set at 2.8-Å resolution (Fig. 4 and fig. S15).

For both states, in the NUC lobe, the RNA–protein interface exhibited a build-up of positive electrostatic potential facilitated by AI-introduced residues K224R, Q227R, T228R, and K245R (Fig. 4, A and B). The RNA was further stabilized by the masked WT residues R232, R238, and K241, which interact with coevolving RNA nucleobases. Additional previously unreported contacts at K256Q and R270K lined the RNA pseudoknot region. Notably, our cryo-EM analysis resolved a TAM-bound conformational intermediate state not previously reported for TnpBs. In this state, the protein and RNA conformations closely resembled the binary complex of ISDra2 TnpB [8BF8 in (*34*)], but the complex was also bound to DNA, containing a single nucleotide pair formed in the guide reRNA–target DNA heteroduplex seed region. Although most of the WT protein-DNA contacts were kept owing to strong coevolution with the TAM motif, the generated N4R substitution stabilized the heteroduplex seed pair together with the conserved "phosphate lock" residue K84 (*45*) (Fig. 4C). Unlike in the ISDra2 structure where the reRNA Stem2 is disordered (*34*), in v7 Stem 2 is structured and bound to the REC lobe.

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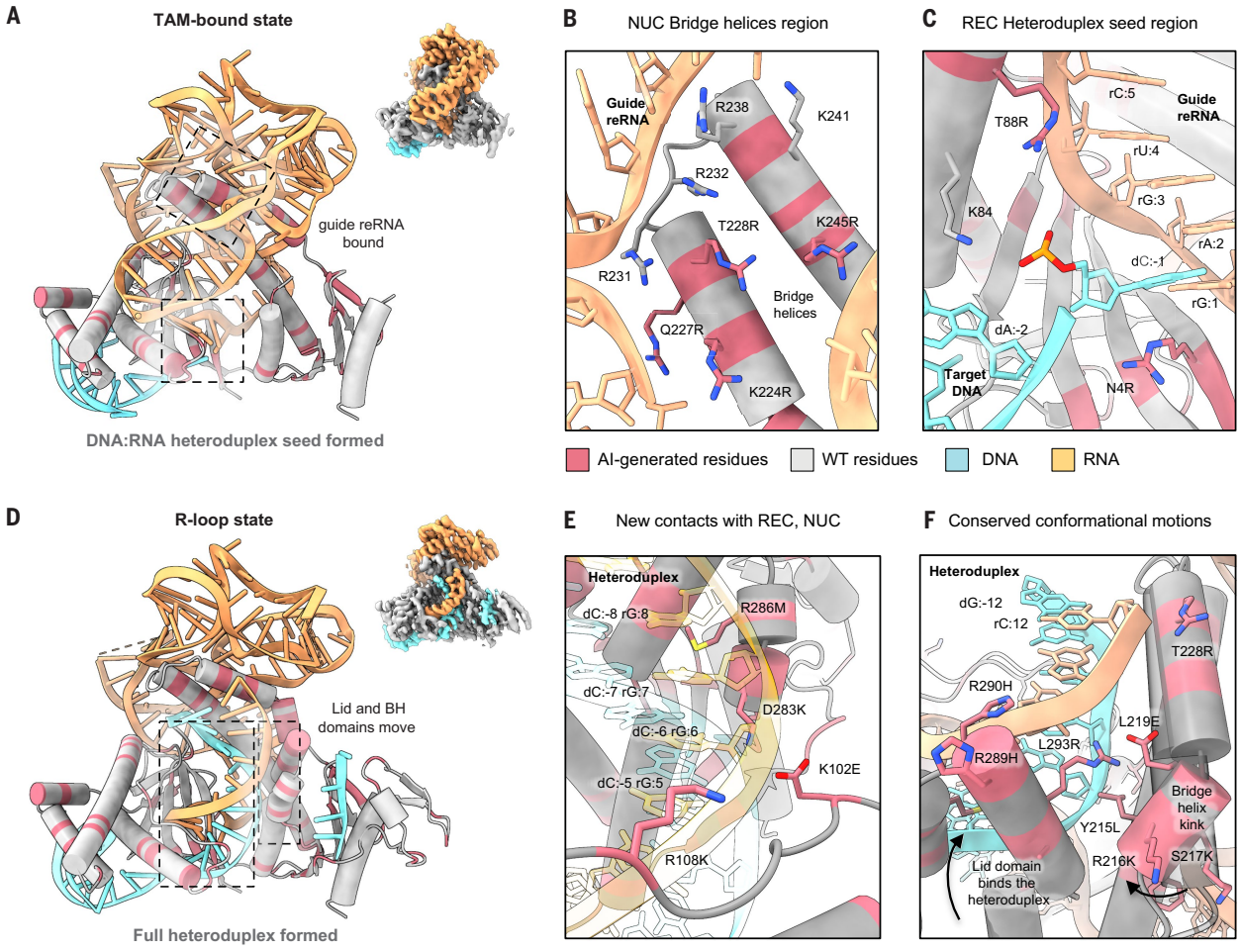


Fig. 4. Cryo-EM structures of the AI-generated variant v7 reveal RNA–DNA interface residues and conserved conformational motions. (**A**) Cryo-EM reconstructions of the TAM-bound (heteroduplex seed formed) and (**D**) R-loop (full heteroduplex) states at 2.8-Å resolution. (**B**) In the NUC lobe, introduced residues (coral) form contacts with the guide reRNA. (**C**) The phosphate lock residue K84 stabilizes the first DNA base pairing with the RNA spacer; putative contacts H4R and T88R appear near the heteroduplex seed. (**E**) Both REC and NUC lobes exhibit additional AI-generated contacts with the RNA:DNA heteroduplex. (**F**) Conserved conformational motions include lid-domain stabilization and bridge-helix kinking mediated by AI-generated residues (L289R-L293R, and Y214F–L219E). Color scheme: AI-generated residues, red; fixed WT residues, gray; RNA, orange; DNA, cyan.

Our R-loop–formed state closely resembled the previously reported ISDra2 TnpB ternary complex [8EXA in (*34*)], with the heteroduplex fully formed along the RNA spacer nucleotides 1 to 12 (Fig. 4D). The REC lobe positions the AI-generated T88R, K102E, R108K, R113K, and R160K residues in contact with the heteroduplex. Further, the lid domain, which caps the substrate-binding cleft, was structured on the heteroduplex through non-natural D283K, R286M, R289H, and L293R contacts (Fig. 4, E and F). Notably, a helical segment largely composed of ESM-IF1–generated residues Y214F, Y215L, R216K, S217K, and L219E preserved the previously observed kink motion that facilitates heteroduplex formation (Fig. 4F) (*34*). Structure-guided reversal mutagenesis supports a functional contribution of a substantial fraction of these AI-introduced residues (fig. S16). Finally, the NUC lobe was fully structured and bound the single-stranded DNA substrate in the RuvC catalytic pocket, which remained lined by residues found in the WT enzyme.

Together, these structures and reversal mutagenesis suggest that the AI-generated residues introduced electrostatic and hydrogen-bonding networks that likely stabilize the RNA–DNA interface in different protein conformational states. The largely AI-generated segment of the protein preserved the native helix kink, consistent with the protein conformational transitions. Finally, our design approach enabled us to capture a previously unobserved TAM-bound state that may not be favored in the WT but is populated in the AI-generated variant.

Discussion

TnpB-family enzymes, which are compact RNA-guided nucleases ancestral to CRISPR-Cas12, are an attractive target for protein design because they couple programmable DNA targeting and a variety of natural functions to a minimal architecture. However, designing such systems has remained limited because sequence-based biological LMs produce proteins that remain close to natural sequences, whereas rational design requires explicit programming of the conformational mechanism.

Unlike LMs that generated proteins that retained WT DNA binding domains with >99% identity to natural homologs (fig. S17) (*12*, *13*), our structure- and evolution-guided design approach created DNA- and RNA-interacting lobes with AI-generated contacts that had 83% and 72% identity to their closest counterparts in nature, respectively. The split-design approach further revealed asymmetric tolerance to sequence change: The DNA-recognition REC lobe was markedly more substitution sensitive than the RNA binding catalytic NUC lobe. We attribute this sensitivity to finely tuned biophysical requirements and stricter evolutionary constraints on TAM search and recognition (*46*), consistent with the need for separate optimization of the REC and NUC lobes. This modular design principle could generalize to other multidomain proteins. Our masking strategy is agnostic to prior experimental structures and is based on evolutionary information. It preserves critical contacts yet permits emergence of contact networks

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that create conformation-specific RNA–DNA interactions across enzymatic states, as supported by our first insight into the cryo-EM structures of the AI-generated RNA-guided nucleases.

Supporting the outcome of our design strategy, recent deep mutational scanning and directed-evolution studies identified that our AI-generated N4R, Q227R, and other substitutions increase ISDra2 TnpB activity (fig. S16) (*47*, *48*). Further, whereas recent AI-guided engineering studies have used inverse-folding models to enhance natural proteins with single-point mutations (*49*, *50*), our distinct strategy integrates phylogenetic conservation with protein–RNA/DNA coevolution, enabling the design of highly divergent non-natural proteins.

The appearance of distinct combinations of DNA–RNA contact residues and their alignment with TnpB's conformational motions raise the possibility that ESM-IF1 may infer fold- and phylogeny-specific protein sequence constraints that give rise to conformational states and residue arrangements compatible with nucleic acid binding, even in the absence of explicit nucleic acid information during training. Refined by evolutionary constraints in our approach, this capability establishes a basis for structure-guided design of other multistate proteins or nucleic acid binders, provided that they are diversely and evenly represented in genomic databases together with their cognate DNA and RNA sequences (*51*–*53*). Furthermore, it has been proposed that proteins function and evolve as ensembles of conformations rather than as single static structures (*54*). In that context, the previously unresolved TAM-bound state that we observe likely represents a distinct kinetic intermediate relative to the WT TnpB and provides a precedent for isolating and studying transient conformations that RNA-guided nucleases may sample in nature and beyond. Establishing the foundation for design of RNA-guided systems and flexible nucleic acid binders with an evolution- and structure-guided model, we envision extending their functions beyond natural sequence constraints through generative biology (*55*) and toward de novo design, expanding the designable protein space.

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SUPPLEMENTARY MATERIALS

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Structure and evolution-guided design of minimal RNA-guided nucleases

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Editor's summary

Enzyme design and engineering are challenging in part because there are few ways to improve catalytic activity but many ways to impair it. Evolution-aware approaches can leverage information from our collection of known natural sequences to guide the generation of diverse engineered enzymes that are more likely to retain function. Skopintsev *et al*. demonstrated engineering of an RNA-guided nuclease using inverse protein-folding models. Screening of candidates nominated by their approach revealed improvement of genome-editing activity. An experimental structure revealed how conformational dynamics are changed in the engineered enzymes, stabilizing the key RNA-DNA interface. —Michael A. Funk

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